

Cybersecurity Incident Report:

Network Traffic Analysis

Part 1: Provide a summary of the problem found in the DNS and ICMP traffic log.

The network traffic analysis shows that the DNS service is affected. The client sends a DNS query using UDP to the DNS server at 203.0.113.2 on port 53. The server responds with an ICMP Destination Unreachable message stating "udp port 53 unreachable." This indicates that the DNS service is not available on UDP port 53, preventing the client from resolving the domain name www.yummyrecipesforme.com. As a result, the browser cannot obtain the destination IP address and cannot load the website.

Part 2: Explain your analysis of the data and provide at least one cause of the incident.

The first observed DNS request occurred at 13:24:32.192571

Many clients reported to our IT team that they couldn't reach the website "www.yummyrecipesforme.com". The page returned the following error message after charging: "Destination port inaccessible"

We first tried to reach the page via our web browser but got the same message error as the clients. We then used the network protocol analyzer tool tcpdump and tried to reach the website again. tcpdump returned that when we have sent UDP packets to the server, we received ICMP packets with the following error message: "udp port 53 unreachable"

Key findings: UDP is being used to send the DNS request, while ICMP is being used to report that UDP port 53 is unreachable. Port 53 is used by DNS. That indicates that no service was listening on UDP port 53 on the DNS server, or that the traffic was being actively rejected.

Likely cause of the incident: the DNS service is unavailable or not listening on UDP port 53. This may be due to the firewall's configuration. Further investigation is required to determine why the DNS service is unavailable.